wholesale_db - mysql | Parts 1-3: build + data verification

```
$ mysql -u root -p < wholesale_distribution_system.sql
Enter password: ********
Welcome to the MySQL monitor.  Commands end with ; or \g.
Server version: 8.0.x MySQL Community Server - GPL

Database changed  (wholesale_db)

mysql> -- CHECK 1: row count of every table
+-------------------+-----------+
| table_name        | row_count |
+-------------------+-----------+
| Suppliers         |         6 |
| Manufacturers     |         7 |
| Products          |        18 |
| Warehouses        |         4 |
| Customers         |        12 |
| Transport_Methods |         4 |
| Supplier_Products |        47 |
| Inventory         |        51 |
| Inventory_Batches |        18 |
| Purchase_Orders   |        21 |
| Order_Items       |        53 |
| Logistics         |        17 |
| Stock_Movements   |      4678 |
+-------------------+-----------+
13 rows in set (0.00 sec)

mysql> -- CHECK 2: ledger vs inventory (expect 0 rows)
Empty set (0.00 sec)

mysql> -- CHECK 3: batch totals vs inventory (expect 0 rows)
Empty set (0.00 sec)

mysql> -- CHECK 4: warehouses over capacity (expect 0 rows)
Empty set (0.00 sec)
```

wholesale_db - mysql | Q1: Low-stock detection

```
mysql> -- Q1: Low-stock detection
```

warehouse	product_name	current_stock	reorder_level	shortfall	pct_of_reorder_level
Lucknow Depot	A4 Copier Paper (500 sheets)	120	220	100	54.5
Kanpur Central Warehouse	Disinfectant Wipes	180	300	120	60.0
Mumbai West Warehouse	Ballpoint Pens (box of 50)	120	200	80	60.0
Kanpur Central Warehouse	Printer Toner Cartridge	38	60	22	63.3
Delhi North Hub	Spiral Notebook 200 pg	150	220	70	68.2
Kanpur Central Warehouse	Office Chair	22	30	8	73.3
Kanpur Central Warehouse	Spiral Notebook 200 pg	380	450	70	84.4
Mumbai West Warehouse	Scientific Calculator	70	80	10	87.5
Lucknow Depot	Whiteboard Markers (12)	190	200	10	95.0

```
9 rows in set (0.00 sec)
```

wholesale_db - mysql | Q1B: Products short in a warehouse

```
mysql> -- Q1B: products short in at least one warehouse
+-------------------------------+---------------------+---------------------------+-----------------------------+
| product_name                  | warehouses_stocking | warehouses_below_reorder | total_stock_all_warehouses |
+-------------------------------+---------------------+---------------------------+-----------------------------+
| Spiral Notebook 200 pg        |                   4 |                         2 |                        1090 |
| A4 Copier Paper (500 sheets)  |                   4 |                         1 |                        2100 |
| Ballpoint Pens (box of 50)    |                   4 |                         1 |                        1610 |
| Disinfectant Wipes            |                   2 |                         1 |                         420 |
| Office Chair                  |                   1 |                         1 |                          22 |
| Printer Toner Cartridge       |                   3 |                         1 |                         138 |
| Scientific Calculator         |                   3 |                         1 |                         420 |
| Whiteboard Markers (12)       |                   3 |                         1 |                        1340 |
+-------------------------------+---------------------+---------------------------+-----------------------------+
8 rows in set (0.00 sec)
```

wholesale_db - mysql | Q2: Best supplier per product

```
mysql> -- Q2: Best supplier per product
```

product_name	best_supplier	unit_price	lead_time_days	quality_score	score
A4 Copier Paper (500 sheets)	Kanpur Office Depot	212.00	2	4.0	92.1
Spiral Notebook 200 pg	Kanpur Office Depot	42.00	2	4.0	93.0
Ballpoint Pens (box of 50)	Kanpur Office Depot	185.00	2	4.0	92.1
Whiteboard Markers (12)	Kanpur Office Depot	170.00	2	4.1	94.1
Heavy Duty Stapler	Kanpur Office Depot	248.00	2	4.0	94.0
Printer Toner Cartridge	Sharma Trading Co.	2100.00	4	4.6	96.5
USB Flash Drive 32GB	Sharma Trading Co.	310.00	4	4.5	95.1
Scientific Calculator	Metro Distributors	548.00	5	4.7	97.3
Hand Sanitizer 500ml	Kanpur Office Depot	138.00	2	4.0	91.7
Floor Cleaner 5L	Kanpur Office Depot	380.00	2	4.0	92.4
Disinfectant Wipes	Lucknow Supply House	180.00	4	3.5	91.0
Instant Coffee 200g	Lucknow Supply House	335.00	4	3.6	91.6
Tea Bags (box of 100)	Lucknow Supply House	152.00	4	3.6	91.6
Bottled Water 1L (case of 12)	Lucknow Supply House	168.00	4	3.5	91.0
Student Desk	Kanpur Office Depot	1880.00	5	4.0	94.0
Office Chair	Kanpur Office Depot	3000.00	5	4.1	94.6
Lab Safety Goggles	Metro Distributors	108.00	5	4.6	95.4
Chemistry Glassware Set	Metro Distributors	1330.00	5	4.7	96.7

```
18 rows in set (0.00 sec)
```

wholesale_db - mysql | Q2B: Supplier leaderboard

```
mysql> -- Q2B: Overall supplier leaderboard
+--------------+-----------------------+-----------------+-----------+---------------------+
| overall_rank | supplier_name         | products_quoted | avg_score | products_where_best |
+--------------+-----------------------+-----------------+-----------+---------------------+
|            1 | Kanpur Office Depot   |               9 |      93.1 |                   9 |
|            2 | Metro Distributors    |              10 |      91.5 |                   3 |
|            3 | Sharma Trading Co.    |               7 |      88.4 |                   2 |
|            4 | Lucknow Supply House  |               8 |      83.8 |                   4 |
|            5 | Bharat Wholesale Mart |               8 |      82.8 |                   0 |
+--------------+-----------------------+-----------------+-----------+---------------------+
5 rows in set (0.00 sec)
```

```
mysql> -- Q3: Stock-out prediction (<= 14 days of cover)
```

warehouse	product_name	current_stock	avg_daily_sales	days_left	predicted_stockout_date
Kanpur Central Warehouse	Office Chair	22	6.0	3.7	2026-11-13
Kanpur Central Warehouse	Disinfectant Wipes	180	36.0	5.0	2026-11-15
Kanpur Central Warehouse	Printer Toner Cartridge	38	6.0	6.3	2026-11-16
Lucknow Depot	A4 Copier Paper (500 sheets)	120	18.0	6.7	2026-11-16
Mumbai West Warehouse	Ballpoint Pens (box of 50)	120	18.0	6.7	2026-11-16
Mumbai West Warehouse	Scientific Calculator	70	9.0	7.8	2026-11-17
Delhi North Hub	Spiral Notebook 200 pg	150	18.0	8.3	2026-11-18
Lucknow Depot	Student Desk	28	3.0	9.3	2026-11-19
Mumbai West Warehouse	Lab Safety Goggles	180	18.0	10.0	2026-11-20
Kanpur Central Warehouse	Spiral Notebook 200 pg	380	36.0	10.6	2026-11-20
Lucknow Depot	Whiteboard Markers (12)	190	18.0	10.6	2026-11-20
Lucknow Depot	Floor Cleaner 5L	95	9.0	10.6	2026-11-20
Mumbai West Warehouse	Chemistry Glassware Set	33	3.0	11.0	2026-11-21
Kanpur Central Warehouse	Scientific Calculator	210	18.0	11.7	2026-11-21
Lucknow Depot	Lab Safety Goggles	210	18.0	11.7	2026-11-21
Delhi North Hub	Student Desk	36	3.0	12.0	2026-11-22
Lucknow Depot	Disinfectant Wipes	240	18.0	13.3	2026-11-23
Lucknow Depot	Chemistry Glassware Set	40	3.0	13.3	2026-11-23
Kanpur Central Warehouse	Tea Bags (box of 100)	500	36.0	13.9	2026-11-23

```
19 rows in set (0.00 sec)
```

wholesale_db - mysql | Q4: Pending purchase orders

```
mysql> -- Q4: Pending purchase orders
+----------+----------------------+--------------------------+------------+-------------+----------+-----------------+---------------+-----------------+--------------+
| order_id | supplier             | destination              | order_date | expected_by | status   | shipment_status | units_pending | order_value_inr | days_overdue |
+----------+----------------------+--------------------------+------------+-------------+----------+-----------------+---------------+-----------------+--------------+
|       15 | Sharma Trading Co.   | Mumbai West Warehouse    | 2026-10-28 | 2026-11-02  | Shipped  | In Transit      |           950 |       191100.00 |            8 |
|       16 | Metro Distributors   | Kanpur Central Warehouse | 2026-10-30 | 2026-11-06  | Shipped  | In Transit      |           840 |       252880.00 |            4 |
|       18 | Lucknow Supply House | Kanpur Central Warehouse | 2026-11-04 | 2026-11-11  | Approved | Not dispatched  |           500 |        75000.00 |            0 |
|       17 | Bharat Wholesale Mart| Delhi North Hub          | 2026-11-01 | 2026-11-12  | Shipped  | In Transit      |            85 |        83100.00 |            0 |
|       19 | Kanpur Office Depot  | Delhi North Hub          | 2026-11-07 | 2026-11-12  | Placed   | Not dispatched  |           180 |        89200.00 |            0 |
|       20 | Metro Distributors   | Delhi North Hub          | 2026-11-09 | 2026-11-16  | Placed   | Not dispatched  |           240 |        91920.00 |            0 |
+----------+----------------------+--------------------------+------------+-------------+----------+-----------------+---------------+-----------------+--------------+
6 rows in set (0.00 sec)
```

wholesale_db - mysql | Q5: Supplier on-time performance

```
mysql> -- Q5: Supplier on-time delivery performance
+----------------------+------------------+----------------+------------+---------------------+
| supplier             | delivered_orders | on_time_orders | on_time_pct | avg_days_vs_promise |
+----------------------+------------------+----------------+------------+---------------------+
| Kanpur Office Depot  |                3 |              3 |      100.0 |                -0.3 |
| Sharma Trading Co.   |                3 |              3 |      100.0 |                -0.7 |
| Metro Distributors   |                3 |              2 |       66.7 |                 0.7 |
| Bharat Wholesale Mart |                2 |              1 |       50.0 |                 1.0 |
| Lucknow Supply House |                2 |              0 |        0.0 |                 3.0 |
| Southern Traders     |                1 |              0 |        0.0 |                 5.0 |
+----------------------+------------------+----------------+------------+---------------------+
6 rows in set (0.00 sec)
```

wholesale_db - mysql | Q6: Warehouse comparison

```
mysql> -- Q6: Warehouse comparison
+--------------------------+-------------+---------------+-----------------+--------------------+---------+
| warehouse                | total_units | product_lines | stock_value_inr | capacity_used_pct | flag    |
+--------------------------+-------------+---------------+-----------------+--------------------+---------+
| Kanpur Central Warehouse |        7780 |            18 |      2765300.00 |               51.9 | HIGHEST |
| Delhi North Hub          |        3311 |            12 |      1133100.00 |               41.4 | -       |
| Mumbai West Warehouse    |        1918 |            11 |       662100.00 |               27.4 | -       |
| Lucknow Depot            |        1623 |            10 |       499900.00 |               27.1 | LOWEST  |
+--------------------------+-------------+---------------+-----------------+--------------------+---------+
4 rows in set (0.00 sec)
```

wholesale_db - mysql | Q7: Replenishment tracking

```
mysql> -- Q7: Replenishment tracking
```

warehouse	product_name	on_hand	reorder_level	qty_on_order	projected_stock	replenishment_status
Delhi North Hub	Spiral Notebook 200 pg	150	220	0	150	ACTION: low and nothing on order
Kanpur Central Warehouse	Printer Toner Cartridge	38	60	0	38	ACTION: low and nothing on order
Kanpur Central Warehouse	Spiral Notebook 200 pg	380	450	0	380	ACTION: low and nothing on order
Lucknow Depot	A4 Copier Paper (500 sheets)	120	220	0	120	ACTION: low and nothing on order
Lucknow Depot	Whiteboard Markers (12)	190	200	0	190	ACTION: low and nothing on order
Mumbai West Warehouse	Scientific Calculator	70	80	0	70	ACTION: low and nothing on order
Kanpur Central Warehouse	Disinfectant Wipes	180	300	200	380	Covered by open order
Kanpur Central Warehouse	Office Chair	22	30	30	52	Covered by open order
Mumbai West Warehouse	Ballpoint Pens (box of 50)	120	200	250	370	Covered by open order
Delhi North Hub	Floor Cleaner 5L	160	80	50	210	Healthy - routine replenishment
Delhi North Hub	Hand Sanitizer 500ml	400	220	100	500	Healthy - routine replenishment
Delhi North Hub	Instant Coffee 200g	220	80	200	420	Healthy - routine replenishment
Delhi North Hub	Printer Toner Cartridge	55	30	25	80	Healthy - routine replenishment
Delhi North Hub	Scientific Calculator	140	80	100	240	Healthy - routine replenishment
Delhi North Hub	Student Desk	36	20	30	66	Healthy - routine replenishment
Kanpur Central Warehouse	Bottled Water 1L (case of 12)	700	400	400	1100	Healthy - routine replenishment
Kanpur Central Warehouse	Hand Sanitizer 500ml	600	450	300	900	Healthy - routine replenishment
Kanpur Central Warehouse	Scientific Calculator	210	150	60	270	Healthy - routine replenishment
Kanpur Central Warehouse	Tea Bags (box of 100)	500	350	350	850	Healthy - routine replenishment
Mumbai West Warehouse	A4 Copier Paper (500 sheets)	360	220	700	1060	Healthy - routine replenishment

20 rows in set (0.00 sec)

wholesale_db - mysql | Q8: Transportation analysis

```
mysql> -- Q8: Transportation analysis
+--------------------+-----------+-----------------+-------------------+------------------+-----------+------------+
| method_name        | shipments | avg_cost_per_km | avg_cost_per_unit | avg_transit_days | cost_rank | speed_rank |
+--------------------+-----------+-----------------+-------------------+------------------+-----------+------------+
| Rail Freight       |         4 |            8.88 |             46.76 |              5.8 |         1 |          4 |
| Road - Mini Truck  |         4 |           13.50 |              1.81 |              1.8 |         2 |          2 |
| Road - Heavy Truck |         5 |           20.00 |             21.94 |              2.6 |         3 |          3 |
| Air Cargo          |         1 |           65.00 |            439.19 |              1.0 |         4 |          1 |
+--------------------+-----------+-----------------+-------------------+------------------+-----------+------------+
4 rows in set (0.00 sec)
```

```
mysql> -- F1: Expiry tracking
```

warehouse	product_name	batch_no	quantity	expiry_date	days_to_expiry	alert_level	value_at_risk
Kanpur Central Warehouse	Tea Bags (box of 100)	TB-2505-A	120	2026-11-05	-5	EXPIRED	26400.00
Lucknow Depot	Disinfectant Wipes	WP-2507-A	240	2026-11-14	4	CRITICAL (<=14 days)	62400.00
Kanpur Central Warehouse	Disinfectant Wipes	WP-2506-A	80	2026-11-18	8	CRITICAL (<=14 days)	20800.00
Kanpur Central Warehouse	Hand Sanitizer 500ml	HS-2406-A	200	2026-11-25	15	WARNING (<=30 days)	38000.00
Delhi North Hub	Hand Sanitizer 500ml	HS-2407-A	150	2026-11-30	20	WARNING (<=30 days)	28500.00
Kanpur Central Warehouse	Instant Coffee 200g	CF-2501-A	100	2026-12-05	25	WARNING (<=30 days)	48000.00
Kanpur Central Warehouse	Bottled Water 1L (case of 12)	WT-2507-A	300	2026-12-20	40	WATCH (<=60 days)	72000.00
Delhi North Hub	Bottled Water 1L (case of 12)	WT-2508-A	250	2026-12-28	48	WATCH (<=60 days)	60000.00

```
8 rows in set (0.00 sec)
```

```
mysql> -- F2: Primary and backup supplier per product
```

product_name	primary_supplier	primary_price	backup_supplier	backup_price	backup_lead_days	backup_price_diff_pct
A4 Copier Paper (500 sheets)	Kanpur Office Depot	212.00	Sharma Trading Co.	208.00	3	-1.9
Ballpoint Pens (box of 50)	Kanpur Office Depot	185.00	Sharma Trading Co.	182.00	3	-1.6
Bottled Water 1L (case of 12)	Lucknow Supply House	168.00	Metro Distributors	176.00	5	4.8
Chemistry Glassware Set	Metro Distributors	1330.00	Bharat Wholesale Mart	1280.00	6	-3.8
Disinfectant Wipes	Lucknow Supply House	180.00	Metro Distributors	188.00	5	4.4
Floor Cleaner 5L	Kanpur Office Depot	380.00	Lucknow Supply House	365.00	4	-3.9
Hand Sanitizer 500ml	Kanpur Office Depot	138.00	Lucknow Supply House	130.00	4	-5.8
Heavy Duty Stapler	Kanpur Office Depot	248.00	Sharma Trading Co.	255.00	3	2.8
Instant Coffee 200g	Lucknow Supply House	335.00	Metro Distributors	350.00	5	4.5
Lab Safety Goggles	Metro Distributors	108.00	Bharat Wholesale Mart	102.00	6	-5.6
Office Chair	Kanpur Office Depot	3000.00	Metro Distributors	3120.00	8	4.0
Printer Toner Cartridge	Sharma Trading Co.	2100.00	Metro Distributors	2060.00	5	-1.9
Scientific Calculator	Metro Distributors	548.00	Bharat Wholesale Mart	535.00	6	-2.4
Spiral Notebook 200 pg	Kanpur Office Depot	42.00	Sharma Trading Co.	44.00	3	4.8
Student Desk	Kanpur Office Depot	1880.00	Metro Distributors	1940.00	8	3.2
Tea Bags (box of 100)	Lucknow Supply House	152.00	Metro Distributors	160.00	5	5.3
USB Flash Drive 32GB	Sharma Trading Co.	310.00	Bharat Wholesale Mart	295.00	7	-4.8
Whiteboard Markers (12)	Kanpur Office Depot	170.00	Sharma Trading Co.	175.00	3	2.9

```
18 rows in set (0.00 sec)
```

```
mysql> START TRANSACTION;  -- receive purchase order 16 ... COMMIT;
Query OK, 0 rows affected

mysql> -- T1: Kanpur stock BEFORE receiving order 16
+-------------------------------+----------+---------------+
| product_name                  | quantity | reorder_level |
+-------------------------------+----------+---------------+
| Scientific Calculator         |      210 |           252 |
| Tea Bags (box of 100)         |      500 |           504 |
| Bottled Water 1L (case of 12) |      700 |           504 |
| Office Chair                  |       22 |            84 |
+-------------------------------+----------+---------------+
4 rows in set (0.00 sec)

mysql> -- T2: Kanpur stock AFTER the COMMIT
+-------------------------------+----------+---------------+
| product_name                  | quantity | reorder_level |
+-------------------------------+----------+---------------+
| Scientific Calculator         |      270 |           252 |
| Tea Bags (box of 100)         |      850 |           504 |
| Bottled Water 1L (case of 12) |     1100 |           504 |
| Office Chair                  |       52 |            84 |
+-------------------------------+----------+---------------+
4 rows in set (0.00 sec)

mysql> -- T3: inside transaction after the mistake
+--------------------+
| kanpur_total_units |
+--------------------+
|                  0 |
+--------------------+
1 row in set (0.00 sec)

mysql> -- T4: after ROLLBACK TO SAVEPOINT
+--------------------+
| kanpur_total_units |
+--------------------+
|               8570 |
+--------------------+
1 row in set (0.00 sec)

mysql> -- T5: after the final ROLLBACK
+--------------------+
| kanpur_total_units |
+--------------------+
|               8620 |
+--------------------+
1 row in set (0.00 sec)

mysql> -- FINAL CHECK: ledger vs inventory (expect 0 rows)
Empty set (0.00 sec)
```